

SENTINEL FUSION

PHASE 1 IDEA SUBMISSION TECHNICAL APPENDIX (03_SUPPORTING_FILES)

This technical appendix delivers comprehensive structural architectures, deterministic processing flows, actor use-case relationship matrix transformations, and granular user interface wireframe schemas for the Sentinel Fusion AI-Native Digital Forensics and Intelligence Platform. All layouts are engineered to ensure explicit operational parameters, human-in-the-loop validation checkpoints, and robust chain-of-custody compliance.

1. System Architecture Diagram

The block topology below defines the structural decoupling of components across Sentinel Fusion. High-throughput data ingestion flows into an asynchronous queuing layer (Redis/Celery), orchestration transitions through specialized LangGraph networks, and persistence models propagate contextually across relational, graph, vector, and indexing engine variants.

FRONTEND PRESENTATION TIER (NEXT.JS 16 / REACT 20)

- User Dashboard Views
- Visualization Layer (React Flow, vis-timeline, MapLibre)
- Forensic Reporting Port
- Secure Session Handshake (Clerk Auth Token validation)

API GATEWAY & CORE ORCHESTRATION TIER (FASTAPI ASYNC CORE)

- Route Request Router
- Audit Log Broadcaster
- Task Job Dispatcher

TRANSACTIONAL OPERATIONS & INGESTION PERSISTENCE / DECOUPLED STORAGE INFRASTRUCTURE TIER

- **Primary Relational DB:** PostgreSQL (Users, Cases)
- **Network Graph DB:** Neo4j (Identity Graphs)
- **Search Index Engine:** OpenSearch (OCR Index)
- **Object Storage Payload:** S3 / Cloudflare R2 / MinIO
- **Vector Database:** Qdrant (RAG / Similarity)
- **Asynchronous Queue:** Redis Cache Cluster (Redis Task Broker Queue Pop!)

AI ANALYTICS & ASYNC PROCESSING CORE (CELERY WORKERS & LANGGRAPH)

- PaddleOCR Engine
- Whisper Speech-to-Text
- Deepfake Media Analyzer
- Florence-2 / Gemma Vision
- Large Language Orchestrator (GPT-5.5 / Llama 4 RAG Core)

2. Evidence Processing Pipeline Flowchart

The deterministic linear processing lifecycle of integrated evidence payloads ensures strict adherence to electronic forensic guidelines, verifying internal authenticity metrics prior to dashboard reporting updates.

STEP 1: Multi-Source Evidence Ingestion Phase

Bulk uploads targeting fragmented packages (.zip archives, raw multimedia files, communication exports, metadata records).

STEP 2: Chain-of-Custody Cryptographic Lockdown

Instant generation of SHA-256 validation signatures matching payload data; record appended to immutable structural server logs.

STEP 3: Payload Structural Partitioning & Routing

Raw binary elements are committed directly to target Object Storage clusters; transaction state mappings are tracked inside PostgreSQL.

STEP 4: Distributed Asynchronous Processing Routing

The primary system API drops structured analytical parameters into the Redis cache broker layer to orchestrate parallel worker tracking.

STEP 5: Parallel Pipeline AI Analytics Execution Ensemble (LangGraph Controlled)

- **ExifTool:** Core file layout and spatial geolocation metrics capture.
- **PaddleOCR:** Text string compilation from visual imagery arrays.
- **OpenAI Whisper:** Speech transcript generation tracking for audio arrays.
- **Deepfake Detection:** Synthetic alteration assessment scores generated.

STEP 6: Cross-Domain Structural Correlation & Intelligence Fusion

Neo4j engine binds communication handles, digital personas, geographic anchors, and system nodes to map structural overlap clusters.

STEP 7: System Core Threat Matrix Assessment Metrics

Dynamic computational hazard scoring models pass compiled operational state arrays to build explainable UI summary readouts.

STEP 8: Interactive Dashboard View Assembly

Operational data elements populate target relational grids, timeline traces, and interactive graphical connection modules.

3. System Use Case Diagrams

The user action matrix defines authorization barriers separating core human investigative roles from structural, automated computational micro-services managed natively by the system processing tier.

User Actor Role	Authorized Platform Use Case	Underlying Architectural Subsystem
Forensic Investigator	Ingest multi-type evidence payloads & seal chain-of-custody.	FastAPI Core, SHA-256 Engine, Object Storage Cluster.
Forensic Investigator	Execute Deepfake structural audit validations on questionable components.	Synthetic Media Detection Pipeline, Visual Confidence Arrays.
Case Officer	Interrogate cross-device node relationship connection charts.	Neo4j Graph Queries, React Flow Presentation Framework.
Case Officer	Audit historical chronological case progression views.	vis-timeline Ingestion, OpenSearch Chrono-sorting Indexes.
Authorized Decision Maker	Evaluate high-risk threat profiling metrics & system text outputs.	LLM Explainer Engine, RAG Database Context Integrator.
Authorized Decision Maker	Generate signed cryptographic compliance evidence export sheets.	Forensic Report Engine, PDF Generation Utilities.
Automated Fusion Engine	Natively scan, read string blocks, and transcribe asset sound archives.	PaddleOCR Workers, Async Whisper Inference Cluster.
Automated Fusion Engine	Map and link identical core identities discovered across disparate data clusters.	LangGraph Orchestrator Network, Qdrant Vector Match Layers.

4. UI Wireframe Visual Blueprints

The structural wireframes below simulate high-fidelity platform operational dashboard environments, displaying interface layout rules, interactive layout paradigms, and system components.

4.1 UI Wireframe 01: Main Investigation Dashboard View

[S] SENTINEL FUSION v1.0 | Case Profile: #2026-NARC-089A

Status: ACTIVE

Investigator: Det. R. Vance

Search: [Search evidence, hashes, entities...]

• Dashboard Overview

Data Ingestion +

Network Graph

Case Timeline

Reports Engine

System Audit Logs

Case Overview & Threat Profile

Risk Level: CRITICAL SCORING (89%)

Core Entities Mapped: 42 Unique System Records

Integrity: SHA-256 Validation Active & Consistent

Real-Time Critical Alerts

[!] HIGH DEEPFAKE PROBABILITY DETECTED: Target audio clip clone probability calculated at 98.4%.

[!] CROSS-CASE SUSPECT LINK FOUND: Target device matches active records inside Case #042B.

Intelligence Visualization Workspace (Active Chrono-Timeline View)

[08:00:23 UTC] Device #1 Uploaded Target Image (Location: Extracted Exif Coordinates)

[08:14:55 UTC] OCR Extraction: Text matching contraband ledger flagged in graphic array

[09:33:10 UTC] Audio processed via Whisper Core: "Drop point coordination parameters locked"

[11:02:14 UTC] Transaction Hash linked to target crypto wallet storage endpoint

Sentinel Fusion • Phase 1 Technical Appendix (03_SUPPORTING_FILES)

Page 5 of 9

4.2 UI Wireframe 02: Evidence Upload & Ingestion Interface

[S] SENTINEL FUSION | Case Payload Management Ingestion Control

[Back to Main]

[Upload Target Manager]

[Batch Processing]

[DRAG & DROP DIGITAL EVIDENCE PAYLOAD FILES HERE]

Supports compressed zip tar archives, mp4, wav, jpg, and raw forensic data dumps

Active Ingestion Processing Queue

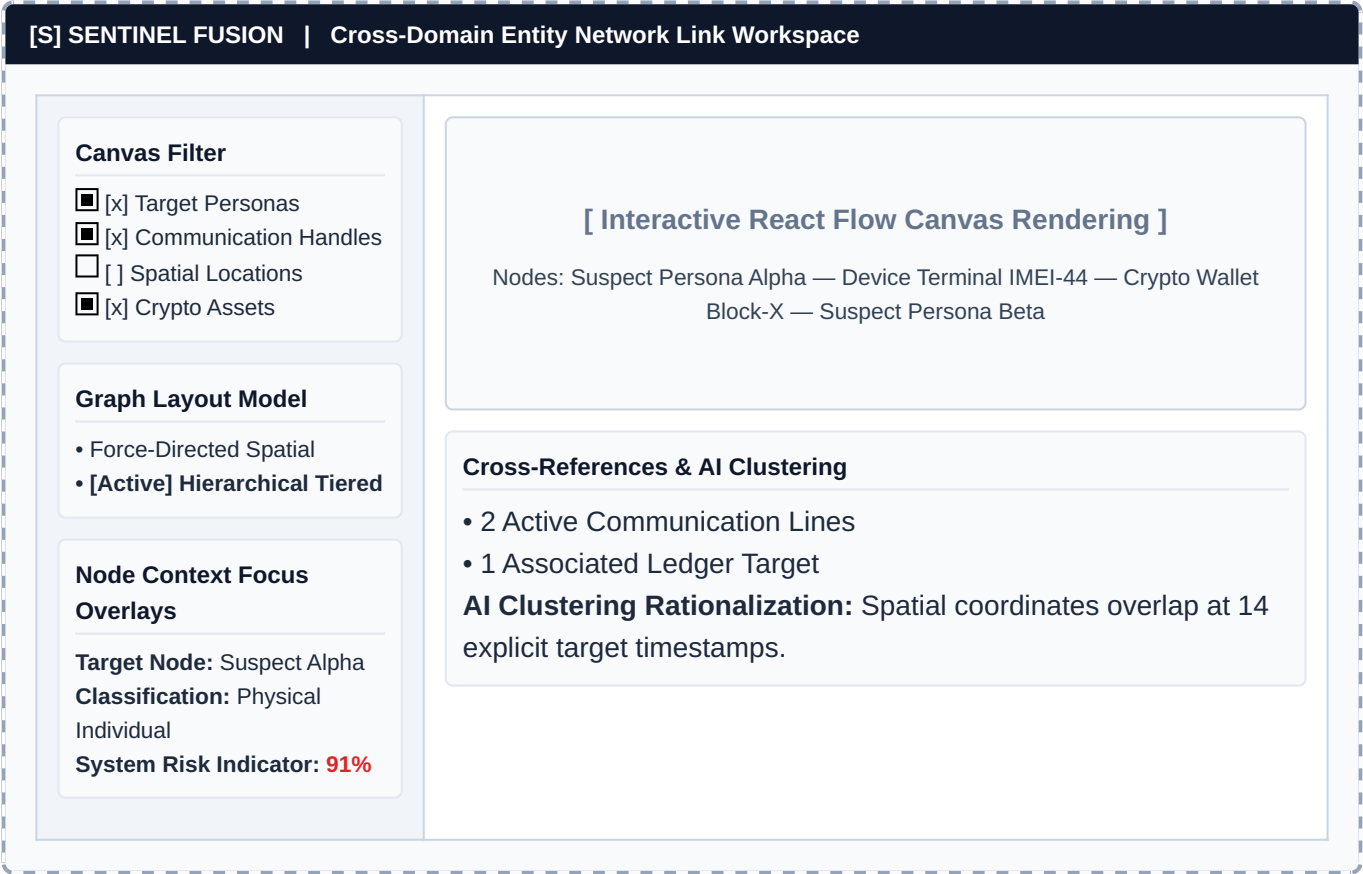
File Target Identifier	Data Size	Pipeline State Status	Generated Cryptographic SHA-256 Hash
device_extraction_dump.zip	4.2 GB	100%	8f3c7b91a2c3d4e5f6a7b8c9d0e1f2a3e92a11bf
intercepted_voicemail.wav	14.8 MB	72%	Computing hash signature...
surveillance_feed_h264.mp4	890.5 MB	Awaiting slot...	Pending pipeline input

Immutable Chain-of-Custody System Log Stream

> [14:22:01 UTC] INGESTION RUNNING: Session initialized under active security credential UID-8902.

> [14:22:05 UTC] CRYPTO LOCK COMPLETE: File device_extraction_dump.zip sealed with immutable structural signature.

4.3 UI Wireframe 03: Interactive Relationship Graph Network View



4.4 UI Wireframe 04: AI Verification & Automated Report Generator

[S] SENTINEL FUSION | Automated Compliance & Reporting Dashboard Export Suite

Included Document Sections: Executive Summary, Chain-of-Custody Log, Deepfake Media Audit, Entity Connection Matrix, System Explainability Index

Target Format: ☒ PDF Forensic Report ☐ Structural JSON Stream ☐ Open Document Format

Automated Report Text Preview Assembly

1. CONSOLIDATED CASE EVIDENCE INGESTION REPORT SUMMARY

The Sentinel Fusion computing platform successfully evaluated 3 core digital forensic intelligence vectors. Computational processing structures flagged localized cellular coordinate overlaps, distinct cross-device connection footprints, and potential structural modification properties inside target sound file tracking headers.

AI SYSTEM TRANSPARENCY, CONFIDENCE INTERVAL & VERIFICATION RATIONALE

The primary engine generated an internal validation baseline of 96.2% across active classification components. Computed threat indicators map strictly to explicit structural variable flags pulled directly from file metadata parameters.

[COMPILER, CRYPTOGRAPHICALLY SIGN & EXPORT FORENSIC COMPLIANCE DOCUMENT]

5. Visual Appendix: System Architecture & Pipeline Diagrams

The following high-fidelity visual diagrams depict the complete structural layout of the Sentinel Fusion platform and its multi-stage data processing pipeline orchestration layer.

